MILITARY SPECIFICATION

ANTENNA AT-840/ARC

1. SCOPE

1.1 This specification covers the requirements for one type of antenna, designated Antenna AT-840/ARC.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

QQ-A-596	Aluminum Alloy Permanent and Semi-Permanent Mold Castings

Military

MIL-C-71	Connectors, "N" For Radio Frequency Cables
MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-E-5422	Environmental Testing, Aircraft Electronic Equipment
MIL-P-7094	Plastic Parts, Aircraft Exterior, General Requirements and Tests for Rain-Erosion, Protection Of
MIL-T-9107	Test Reports, Preparation Of
MIL-P-17555	Preparation For Delivery Of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) And Associated Repair Parts

STANDARDS

Military

MIL-STD-130	Identification Marking of U. S. Military Property

DRAWINGS

Air Force

58D12573	Antenna AT-840/ARC - Outline Dimensions Of

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(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as required by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for preproduction testing.

3.2 General Specification.- The requirements of MIL-E-5400 for class 2 equipment apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.2.1 Connectors.- The antenna connector shall be designed to accommodate Plug UG-21D/U in accordance with MIL-C-71.

3.2.2 Aluminum Castings.- Aluminum castings shall be of a nonporous variety and shall be in accordance with QQ-A-596.

3.2.3 Service Conditions (Environmental).- The mechanical and electrical characteristics of the antenna shall meet the requirements stated herein when tested in accordance with MIL-E-5422.

3.2.3.1 High Temperature.- The antenna shall be capable of continuous operation at temperatures up to 260°F without affecting the electrical or mechanical performance.

3.2.4 Dimensions.- Envelope dimensions and mounting dimensions of the antenna shall be specified in Drawing 58D12573.

3.2.5 Sealing.- All cavities in the antenna normally not occupied by solid structural dielectrics shall be filled with a high-curing-temperature foam dielectric to prevent collection of liquids. Openings used for foaming shall be securely covered with suitable seals to prevent entry of moisture or other liquids. Other techniques for sealing may be used subject to meeting the electrical, mechanical, and environmental requirements specified herein.

3.2.6 Side Thrust Strength.- The antenna shall be designed to withstand side thrust load of 8 pounds per square inch of lateral area.

3.2.7 Erosion Resistance.- All portions of the antenna which protrude more than three-fourths of an inch into the airstream shall be metallic or shall have metallic-covered leading edges to prevent rain erosion. All other plastic surfaces shall be protected from rain erosion in accordance with MIL-P-7094.

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3.3 Performance

3.3.1 Frequency.- The antenna shall provide performance over the radio frequency range of 225 to 400 megacycles per second.

3.3.2 Polarization.- The antenna shall transmit or receive vertically polarized signals. Transmission or reception of horizontally polarized signals shall be at least 15 db down from that of the vertically polarized signals when tested according to 4.3.4.6 herein. The antenna shall be so designed that the ratio of maximum gain to minimum gain for the vertical polarized signal shall be less than 2 db when tested according to 4.3.4.6.

3.3.3 Impedance.- The impedance of the antenna measured at the connection of the transmission line shall be such that the voltage standing wave ratio (VSWR) produced on Radio Frequency Cable RG-8/U does not exceed 2.5 to 1 between 225 and 299 megacycles per second and 2.0 to 1 between 300 and 400 megacycles per second when tested in accordance with 4.3.4.5.

3.3.4 Radiation Pattern.- The antenna shall be designed for omni-directional radiation pattern coverage. The vertical polarization pattern shall be free of deep nulls when tested according to 4.3.4.6 herein.

3.3.5 Radiation Power.- The antenna shall be capable of handling an average power of 100 watts when subjected to any combination of the environmental conditions specified herein.

3.3.6 Efficiency.- The antenna assembly, including matching network, if used, shall be constructed of materials which are classified as good conductors or good dielectrics. No resistive components shall be used.

3.3.7 Radiation Efficiency.- The antenna shall have a gain within 1 db of a matched quarter-wave stub at any test frequency between 225 and 400 megacycles per second, when tested according to 4.3.4.7.

3.4 Weight.- The weight of the antenna shall not exceed 24 ounces.

3.5 Identification of Product.- Assemblies, subassemblies, and parts shall be marked in accordance with MIL-STD-130, and shall contain the information and be located as shown on Drawing 58D12573.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests.- The inspection and testing of the antenna shall be classified as follows:

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a. Preproduction tests . . . . . See 4.3

b. Acceptance tests . . . . . See 4.4.

4.2 Test Conditions.- Unless otherwise specified, the antennas shall be tested under the following conditions:

Temperature:	Plus 15°C to plus 35°C
Pressure:	Normal atmospheric
Humidity:	Prevailing ambient up to 90 percent relative humidity.

4.3 Preproduction Tests

4.3.1 Preproduction Test Samples.- When specified in the contract, a minimum of three antennas shall be subjected by the manufacturer to the preproduction tests and the acceptance tests specified herein.

4.3.1.1 Samples For Procuring Activity.- Three antennas shall be submitted to the procuring activity at the time the manufacturer submits the preproduction test report for approval. These antennas shall be used by the Government for:

a. A review of the electrical characteristics and mechanical construction of the antenna.

b. The performance of any test deemed necessary after a review of the manufacturer's report of tests on his preproduction sample.

c. Government reference purposes.

4.3.2 Preproduction Test Program.- Prior to any formal preproduction testing of the sample antennas, the manufacturer shall submit a draft of his proposed test program for approval to the procuring activity. The test program shall include the procedure and equipment used in performing the preproduction tests specified herein.

4.3.3 Preproduction Test Report.- After completion of the preproduction tests, three copies of the complete report shall be furnished the procuring activity. The test report shall be in accordance with MIL-T-9107 and shall include the name and grade of all materials used in the antenna.

4.3.4 Tests

4.3.4.1 Environmental Test.- Environmental tests shall be performed by an approved testing laboratory. The sample antennas shall be

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subjected to environmental tests in accordance with the following procedures of MIL-E-5422:

Temperature test	Step 2 and step 6
Humidity test	Step 1 thru step 6
Vibration test	Step 1 thru step 4
Shock test	
Salt spray test	

4.3.4.2 High Temperature Test.- The sample antennas shall be placed within a chamber and the internal temperature shall be raised to 260°F with an internal relative humidity of not more than 5 percent. The antennas shall remain in this environment for 4 hours at the end of which time the antennas shall be removed from the chamber and immediately measured for VSWR.

4.3.4.3 Side Thrust Strength Test.- A distributed load equal to the lateral area (one side only) of the antenna in square inches multiplied by 8 pounds shall be applied at an angle of 90° to the normal direction of flight for a period of 30 minutes. At the end of this period the antenna shall be inspected for possible failure or evidences of cold flow or cracks. The seal test of 4.3.4.4 shall be repeated after the above test to determine possible rupture of seals.

4.3.4.4 Seal Test.- All samples shall be subjected to a seal test by the altitude chamber method. The antenna shall be fully submerged in a water-filled, transparent container which shall be located in the altitude chamber in order to permit visual detection of leaks. The chamber shall be evacuated to a pressure of 1.32 inches of mercury simulating an altitude of 70,000 feet. A continuous flow of bubbles from the antenna shall be cause for rejection.

4.3.4.5 Voltage Standing Wave Ratios (VSWR) Test.- The VSWR of the antenna shall be measured with the antenna installed in the center of an 8-foot-square ground plane using conventional slotted line or radio frequency techniques. The impedence of the antenna shall be measured over the frequency range of 225 to 400 megacycles per second at intervals no greater than 25 mc. VSWR measurements of the antenna shall be made before and after each environmental test as well as during the temperature extremes.

4.3.4.6 Polarization Test.- The antenna under test, when installed in the center of an 8-foot-diameter ground plane for any vertical angle from 0° to +30° above the ground plane and at any angle in azimuth,

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shall have a ratio of vertical polarization to horizontal polarization no less than 15 db. The antenna under tests, when installed on the 8-foot ground plane, shall have omnidirectional pattern coverage and the pattern shall be circular within 2 db for vertical polarization.

4.3.4.7 Radiation Efficiency.- The antenna under test, when installed in the center of an 8-foot diameter ground plane, shall have a gain within 1 db when compared to a matched quarter-wave stub cut to the test frequency. The gain measurements shall be made at the horizon (ground plane) to vertical angles of +30°, above the ground plane, and at any azumith angle. The comparisons shall be made at the same vertical and azimuth angle.

4.3.4.8 Radio Frequency Power Handling Test.- One hundred watts radio-frequency power within the frequency range of the antenna specified herein shall be applied for 20 minutes at room temperature with the antenna mounted in the same ground plane as used in the VSWR test. This test shall be performed immediately following the humidity test of 4.3.4.1. There shall be no evidence of excessive heating of the dielectrics or radiating elements after application of the radio-frequency power.

4.4 Acceptance Test.- Acceptance tests shall consist of:

a. Individual tests

b. Sampling test.

4.4.1 Individual Tests.- Each antenna shall be subjected to the following tests. The right is reserved to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification.

4.4.1.1 Visual Inspection.- Antennas shall be visually inspected to insure that materials, workmanship, and assembly of the antennas meet the requirements of this specification.

4.4.1.2 Voltage Standing Wave Ratio of Antenna.- The VSWR measurement shall be made at a minimum of four frequencies spaced over the band where the VSWR readings have been determined previously to be at a maximum.

4.4.1.3 Seal Test.- Seal tests shall be performed on each antenna in the manner prescribed in 4.3.4.4.

4.4.2 Sampling Test.- One antenna shall be selected at random from every 300 or fraction thereof produced and subjected to the preproduction tests specified herein.

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4.4.2.1 Acceptance Criteria.- When one or more items fail to meet the specification, acceptance of all items will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

5. PREPARATION FOR DELIVERY

5.1 General.- The antenna shall be preserved, packaged, packed, and marked in accordance with MIL-P-17555, method III.

5.2 Shipment Marking.- Shipment marking nomenclature shall be Antenna AT-840/ARC.

6. NOTES

6.1 Intended Use.- This antenna is a radiation element having an essentially omnidirectional vertically polarized radiation pattern intended for use with UHF communication sets.

6.2 Ordering Data.- Procurement documents should state the levels of preservation, packaging, and packing required and the conditions for the following:

6.2.1 Preproduction Tests.- It is expected that the contract or purchase order will specify that a maximum of three antennas will be required as preproduction samples to be subjected to the preproduction tests by the contractor to determine compliance with the requirements of this specification and that these antennas will be required as preproduction samples to be submitted to the procuring activity.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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